

Analysis of running inference on sample.mp4 using ssd_plus_mobilenet_v2_l3f_20260807_225848

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Situation overview & terminology

This analysis aims to examine whether `ssd_plus_mobilenet_v2_l3f_20260807_225848` is usable as the CV-model for the l3f project. I ran an experiment where I took a representative of the real use case. I took a sample of 200 frames from the video. Each of the frames have been run through NN inference. Thus I have an output of 200 frames and N detected objects. Using a self-developed annotation tool I then went through all detected objects and gave each detection one of the following three classifications:

- TP (True Positive): An object is detected and classified correctly
- GHOST: The NN detects an object where there isn't one (sees a ghost)
- MIS-ID (Misidentification): The NN detects an object but misclassifies it (an orange is seen when an apple is actually in the frame; the location of the detected orange matches the actual apple)

This data is saved in a .jsonl format. Each detection also has a NN-defined confidence score. My current hypothesis is that a higher score means a higher chance that the detection was correct (TP). If this is true, we could theoretically choose a threshold which would cut off the incorrectly detected objects, thus make our NN usable in the l3f application. Below we run an analysis to find this threshold.

Setup

Render this document by running `Rscript -e "rmarkdown::render('analysis.Rmd')" From data_and_training/data/myvideotest/sample_frames/metadata/`

Libraries:

```
# for loading data into analysis
library(jsonlite)

# For data manipulation
suppressPackageStartupMessages(library(dplyr)) # <- Don't print warning on import
library(tidyr)

# For plotting & rendering
library(ggplot2)
library(scales)
library(knitr)
```

Data input:

```
input_file <- "results_annotated.jsonl"

raw_data <- stream_in(file(input_file), verbose = FALSE)

length(raw_data)
```

```
## [1] 2
```

Data flattening:

```
detections <- raw_data %>%
  mutate(
    frame = image
  ) %>%
  select(frame, detections) %>%
  unnest(detections) %>%
  mutate(
    annotation = factor(
      annotation,
      levels = c("TP", "MIS-ID", "GHOST")
    )
  ) %>%
  select(
    frame,
    name,
    score,
    annotation
  ) %>%
  rename(
    class = name
  )

print(as.data.frame(head(detections)), row.names = FALSE)
```

##	frame	class	score	annotation
##	hv_0001.jpg	orange (fruit)	0.2192693	MIS-ID
##	hv_0001.jpg	orange (fruit)	0.1549186	MIS-ID
##	hv_0001.jpg	apple	0.1459181	GHOST
##	hv_0001.jpg	knife	0.1244884	GHOST
##	hv_0001.jpg	knife	0.1117359	GHOST
##	hv_0001.jpg	bowl	0.1052758	TP

Tables

Score ranges & TP rate:

```
score_breaks <- seq(0, 1, by = 0.05)

score_range_stats <- detections %>%
  mutate(
    score_range = cut(
      score,
      breaks = score_breaks,
      include.lowest = TRUE,
      right = FALSE
    )
  ) %>%
  group_by(score_range) %>%
  summarise(
    `n(TP)` = sum(annotation == "TP"),
    `n(MIS-ID)` = sum(annotation == "MIS-ID"),
    `n(GHOST)` = sum(annotation == "GHOST"),
    TP_rate = sum(annotation == "TP") / n(),
  )
```

```

    .groups = "drop"
  )

print(score_range_stats %>%
  mutate(
    TP_rate = percent(TP_rate, accuracy = 0.1)
  ) %>%
  as.data.frame(row.names = NULL),
  row.names = FALSE
)

```

```

##  score_range n(TP) n(MIS-ID) n(GHOST) TP_rate
##  [0.1,0.15)   90      194      191   18.9%
##  [0.15,0.2)   57       86       86   24.9%
##  [0.2,0.25)   32       54       56   22.5%
##  [0.25,0.3)   27       42       32   26.7%
##  [0.3,0.35)   19       24       23   28.8%
##  [0.35,0.4)   31       21        8   51.7%
##  [0.4,0.45)   15       10       10   42.9%
##  [0.45,0.5)   21        9        7   56.8%
##  [0.5,0.55)   16        6        3   64.0%
##  [0.55,0.6)   13        3        3   68.4%
##  [0.6,0.65)   11        2        0   84.6%
##  [0.65,0.7)    6        2        1   66.7%
##  [0.7,0.75)    4        1        1   66.7%
##  [0.75,0.8)    3        0        0  100.0%
##  [0.8,0.85)    1        0        0  100.0%

```

Thresholds & Precision:

```

highest_score <- max(detections$score, na.rm = TRUE)
# ceiling would round up to an integer. We want to round up to the next hundredth
# (seuraavaan sadasosaan)
threshold_max <- ceiling(highest_score * 100) / 100 + 0.01
thresholds <- seq(0.10, threshold_max, by = 0.01)

all_tp <- sum(detections$annotation == "TP")

threshold_stats <- lapply(thresholds, function(t) {

  accepted <- detections %>%
    filter(score >= t)

  n_accepted <- nrow(accepted)

  n_tp <- sum(accepted$annotation == "TP")
  n_mis_id <- sum(accepted$annotation == "MIS-ID")
  n_ghost <- sum(accepted$annotation == "GHOST")

  precision <- if (n_accepted > 0) {
    n_tp / n_accepted
  } else {
    NA_real_
  }
})

```

```

tp_retention <- if (all_tp > 0) {
  n_tp / all_tp
} else {
  NA_real_
}

data.frame(
  threshold = t,
  n_accepted = n_accepted,
  n_tp = n_tp,
  n_mis_id = n_mis_id,
  n_ghost = n_ghost,
  precision = precision,
  tp_retention = tp_retention
)
}) %>%
  bind_rows()

threshold_stats %>%
  transmute(
    threshold = sprintf("%.2f", threshold),
    `n(accepted)` = n_accepted,
    `n(TP)` = n_tp,
    `n(MIS-ID)` = n_mis_id,
    `n(GHOST)` = n_ghost,
    `precision` = percent(precision, accuracy = 0.1)
  )

```

##	threshold	n(accepted)	n(TP)	n(MIS-ID)	n(GHOST)	precision
## 1	0.10	1221	346	454	421	28.3%
## 2	0.11	1109	327	405	377	29.5%
## 3	0.12	990	305	362	323	30.8%
## 4	0.13	892	281	319	292	31.5%
## 5	0.14	807	274	278	255	34.0%
## 6	0.15	746	256	260	230	34.3%
## 7	0.16	688	240	241	207	34.9%
## 8	0.17	644	230	224	190	35.7%
## 9	0.18	599	220	204	175	36.7%
## 10	0.19	560	210	193	157	37.5%
## 11	0.20	517	199	174	144	38.5%
## 12	0.21	484	192	163	129	39.7%
## 13	0.22	446	185	148	113	41.5%
## 14	0.23	424	179	138	107	42.2%
## 15	0.24	397	170	130	97	42.8%
## 16	0.25	375	167	120	88	44.5%
## 17	0.26	351	160	109	82	45.6%
## 18	0.27	332	156	103	73	47.0%
## 19	0.28	312	152	93	67	48.7%
## 20	0.29	298	148	88	62	49.7%
## 21	0.30	274	140	78	56	51.1%
## 22	0.31	264	139	74	51	52.7%
## 23	0.32	246	132	68	46	53.7%
## 24	0.33	229	130	61	38	56.8%
## 25	0.34	215	125	55	35	58.1%

## 26	0.35	208	121	54	33	58.2%
## 27	0.36	195	114	50	31	58.5%
## 28	0.37	184	110	44	30	59.8%
## 29	0.38	168	101	39	28	60.1%
## 30	0.39	163	98	38	27	60.1%
## 31	0.40	148	90	33	25	60.8%
## 32	0.41	142	87	33	22	61.3%
## 33	0.42	133	83	29	21	62.4%
## 34	0.43	126	81	27	18	64.3%
## 35	0.44	121	77	26	18	63.6%
## 36	0.45	113	75	23	15	66.4%
## 37	0.46	110	72	23	15	65.5%
## 38	0.47	103	69	21	13	67.0%
## 39	0.48	94	64	18	12	68.1%
## 40	0.49	87	60	16	11	69.0%
## 41	0.50	76	54	14	8	71.1%
## 42	0.51	65	48	10	7	73.8%
## 43	0.52	63	46	10	7	73.0%
## 44	0.53	56	40	9	7	71.4%
## 45	0.54	53	40	8	5	75.5%
## 46	0.55	51	38	8	5	74.5%
## 47	0.56	48	36	8	4	75.0%
## 48	0.57	42	32	7	3	76.2%
## 49	0.58	39	30	7	2	76.9%
## 50	0.59	36	28	6	2	77.8%
## 51	0.60	32	25	5	2	78.1%
## 52	0.61	29	24	3	2	82.8%
## 53	0.62	26	21	3	2	80.8%
## 54	0.63	22	17	3	2	77.3%
## 55	0.64	22	17	3	2	77.3%
## 56	0.65	19	14	3	2	73.7%
## 57	0.66	15	11	2	2	73.3%
## 58	0.67	13	9	2	2	69.2%
## 59	0.68	12	9	2	1	75.0%
## 60	0.69	12	9	2	1	75.0%
## 61	0.70	10	8	1	1	80.0%
## 62	0.71	7	5	1	1	71.4%
## 63	0.72	6	5	1	0	83.3%
## 64	0.73	5	5	0	0	100.0%
## 65	0.74	5	5	0	0	100.0%
## 66	0.75	4	4	0	0	100.0%
## 67	0.76	4	4	0	0	100.0%
## 68	0.77	2	2	0	0	100.0%
## 69	0.78	2	2	0	0	100.0%
## 70	0.79	1	1	0	0	100.0%
## 71	0.80	1	1	0	0	100.0%
## 72	0.81	1	1	0	0	100.0%
## 73	0.82	1	1	0	0	100.0%
## 74	0.83	1	1	0	0	100.0%
## 75	0.84	0	0	0	0	<NA>
## 76	0.85	0	0	0	0	<NA>

Thresholds & Precision with Portion of all TPs accepted:

```

threshold_stats %>%
  transmute(
    threshold = sprintf("%.2f", threshold),
    precision = percent(precision, accuracy = 0.1),
    `portion of all TPs accepted` =
      percent(tp_retention, accuracy = 0.1)
  )

```

##	threshold	precision	portion of all TPs accepted
## 1	0.10	28.3%	100.0%
## 2	0.11	29.5%	94.5%
## 3	0.12	30.8%	88.2%
## 4	0.13	31.5%	81.2%
## 5	0.14	34.0%	79.2%
## 6	0.15	34.3%	74.0%
## 7	0.16	34.9%	69.4%
## 8	0.17	35.7%	66.5%
## 9	0.18	36.7%	63.6%
## 10	0.19	37.5%	60.7%
## 11	0.20	38.5%	57.5%
## 12	0.21	39.7%	55.5%
## 13	0.22	41.5%	53.5%
## 14	0.23	42.2%	51.7%
## 15	0.24	42.8%	49.1%
## 16	0.25	44.5%	48.3%
## 17	0.26	45.6%	46.2%
## 18	0.27	47.0%	45.1%
## 19	0.28	48.7%	43.9%
## 20	0.29	49.7%	42.8%
## 21	0.30	51.1%	40.5%
## 22	0.31	52.7%	40.2%
## 23	0.32	53.7%	38.2%
## 24	0.33	56.8%	37.6%
## 25	0.34	58.1%	36.1%
## 26	0.35	58.2%	35.0%
## 27	0.36	58.5%	32.9%
## 28	0.37	59.8%	31.8%
## 29	0.38	60.1%	29.2%
## 30	0.39	60.1%	28.3%
## 31	0.40	60.8%	26.0%
## 32	0.41	61.3%	25.1%
## 33	0.42	62.4%	24.0%
## 34	0.43	64.3%	23.4%
## 35	0.44	63.6%	22.3%
## 36	0.45	66.4%	21.7%
## 37	0.46	65.5%	20.8%
## 38	0.47	67.0%	19.9%
## 39	0.48	68.1%	18.5%
## 40	0.49	69.0%	17.3%
## 41	0.50	71.1%	15.6%
## 42	0.51	73.8%	13.9%
## 43	0.52	73.0%	13.3%
## 44	0.53	71.4%	11.6%
## 45	0.54	75.5%	11.6%

## 46	0.55	74.5%	11.0%
## 47	0.56	75.0%	10.4%
## 48	0.57	76.2%	9.2%
## 49	0.58	76.9%	8.7%
## 50	0.59	77.8%	8.1%
## 51	0.60	78.1%	7.2%
## 52	0.61	82.8%	6.9%
## 53	0.62	80.8%	6.1%
## 54	0.63	77.3%	4.9%
## 55	0.64	77.3%	4.9%
## 56	0.65	73.7%	4.0%
## 57	0.66	73.3%	3.2%
## 58	0.67	69.2%	2.6%
## 59	0.68	75.0%	2.6%
## 60	0.69	75.0%	2.6%
## 61	0.70	80.0%	2.3%
## 62	0.71	71.4%	1.4%
## 63	0.72	83.3%	1.4%
## 64	0.73	100.0%	1.4%
## 65	0.74	100.0%	1.4%
## 66	0.75	100.0%	1.2%
## 67	0.76	100.0%	1.2%
## 68	0.77	100.0%	0.6%
## 69	0.78	100.0%	0.6%
## 70	0.79	100.0%	0.3%
## 71	0.80	100.0%	0.3%
## 72	0.81	100.0%	0.3%
## 73	0.82	100.0%	0.3%
## 74	0.83	100.0%	0.3%
## 75	0.84	<NA>	0.0%
## 76	0.85	<NA>	0.0%

Per-class medians:

```
class_medians <- detections %>%
  group_by(class) %>%
  summarise(
    `median(TP)` =
      if (any(annotation == "TP"))
        median(score[annotation == "TP"])
      else
        NA_real_,

    `median(MIS-ID)` =
      if (any(annotation == "MIS-ID"))
        median(score[annotation == "MIS-ID"])
      else
        NA_real_,

    `median(GHOST)` =
      if (any(annotation == "GHOST"))
        median(score[annotation == "GHOST"])
      else
        NA_real_,
```

```

    .groups = "drop"
  )

print(as.data.frame(class_medians), row.names = FALSE)

##           class median(TP) median(MIS-ID) median(GHOST)
##      apple  0.1727711      0.1985055      0.1519462
##     banana  0.3248693      0.1831373      0.1631531
##      bowl   0.3389767      0.2166517      0.1440336
##      knife  0.2770019      0.1799821      0.1663212
## orange (fruit) 0.2777624      0.1274895      0.1811838
##      spoon           NA      0.1892162      0.1589111

```

Mean of per class median TPs, MIS-IDs, GHOSTs:

```

class_median_means <- data.frame(
  outcome = c("TP", "MIS-ID", "GHOST"),
  mean_of_class_medians = c(
    mean(class_medians$`median(TP)` , na.rm = TRUE),
    mean(class_medians$`median(MIS-ID)` , na.rm = TRUE),
    mean(class_medians$`median(GHOST)` , na.rm = TRUE)
  )
)

class_median_means

##   outcome mean_of_class_medians
## 1      TP          0.2782763
## 2 MIS-ID          0.1824970
## 3  GHOST          0.1609248

```

Per-frame statistics:

```

all_frames <- raw_data %>%
  transmute(frame = image)

frame_stats <- all_frames %>%
  left_join(
    detections %>%
      group_by(frame) %>%
      summarise(
        `n(TP)` = sum(annotation == "TP"),
        `n(MIS-ID)` = sum(annotation == "MIS-ID"),
        `n(GHOST)` = sum(annotation == "GHOST"),
        n_detections = n(),
        .groups = "drop"
      ),
    by = "frame"
  ) %>%
  mutate(
    `n(TP)` = coalesce(`n(TP)`, 0L),
    `n(MIS-ID)` = coalesce(`n(MIS-ID)`, 0L),
    `n(GHOST)` = coalesce(`n(GHOST)`, 0L),
    n_detections = coalesce(n_detections, 0L),

    `TP rate` = if_else(

```

```

    n_detections > 0,
    `n(TP)` / n_detections,
    NA_real_
  )
) %>%
select(
  frame,
  `n(TP)`,
  `n(MIS-ID)`,
  `n(GHOST)`,
  `TP rate`
)

frame_stats %>%
mutate(
  `TP rate` = ifelse(
    is.na(`TP rate`),
    "N/A",
    percent(`TP rate`, accuracy = 0.1)
  )
)

```

##	frame	n(TP)	n(MIS-ID)	n(GHOST)	TP rate
## 1	hv_0001.jpg	1	2	4	14.3%
## 2	hv_0002.jpg	0	0	2	0.0%
## 3	hv_0003.jpg	0	0	0	N/A
## 4	hv_0004.jpg	6	0	2	75.0%
## 5	hv_0005.jpg	1	7	0	12.5%
## 6	hv_0006.jpg	2	9	0	18.2%
## 7	hv_0007.jpg	2	2	1	40.0%
## 8	hv_0008.jpg	1	3	1	20.0%
## 9	hv_0009.jpg	2	19	1	9.1%
## 10	hv_0010.jpg	4	31	0	11.4%
## 11	hv_0011.jpg	3	23	0	11.5%
## 12	hv_0012.jpg	8	7	4	42.1%
## 13	hv_0013.jpg	16	6	1	69.6%
## 14	hv_0014.jpg	1	10	2	7.7%
## 15	hv_0015.jpg	0	0	4	0.0%
## 16	hv_0016.jpg	1	0	0	100.0%
## 17	hv_0017.jpg	0	0	1	0.0%
## 18	hv_0018.jpg	0	0	2	0.0%
## 19	hv_0019.jpg	1	0	1	50.0%
## 20	hv_0020.jpg	1	0	9	10.0%
## 21	hv_0021.jpg	0	2	1	0.0%
## 22	hv_0022.jpg	0	6	7	0.0%
## 23	hv_0023.jpg	0	0	2	0.0%
## 24	hv_0024.jpg	2	0	1	66.7%
## 25	hv_0025.jpg	1	0	3	25.0%
## 26	hv_0026.jpg	1	0	3	25.0%
## 27	hv_0027.jpg	1	0	1	50.0%
## 28	hv_0028.jpg	0	0	2	0.0%
## 29	hv_0029.jpg	1	1	0	50.0%
## 30	hv_0030.jpg	0	1	2	0.0%
## 31	hv_0031.jpg	2	2	1	40.0%

## 32	hv_0032.jpg	0	2	1	0.0%
## 33	hv_0033.jpg	5	0	1	83.3%
## 34	hv_0034.jpg	2	5	2	22.2%
## 35	hv_0035.jpg	1	5	1	14.3%
## 36	hv_0036.jpg	3	1	3	42.9%
## 37	hv_0037.jpg	2	1	3	33.3%
## 38	hv_0038.jpg	2	3	2	28.6%
## 39	hv_0039.jpg	2	2	2	33.3%
## 40	hv_0040.jpg	3	1	1	60.0%
## 41	hv_0041.jpg	2	2	0	50.0%
## 42	hv_0042.jpg	4	1	1	66.7%
## 43	hv_0043.jpg	2	2	3	28.6%
## 44	hv_0044.jpg	7	1	1	77.8%
## 45	hv_0045.jpg	2	2	4	25.0%
## 46	hv_0046.jpg	1	0	1	50.0%
## 47	hv_0047.jpg	3	1	4	37.5%
## 48	hv_0048.jpg	1	3	3	14.3%
## 49	hv_0049.jpg	1	2	0	33.3%
## 50	hv_0050.jpg	2	2	0	50.0%
## 51	hv_0051.jpg	3	0	3	50.0%
## 52	hv_0052.jpg	0	1	0	0.0%
## 53	hv_0053.jpg	0	1	1	0.0%
## 54	hv_0054.jpg	0	1	1	0.0%
## 55	hv_0055.jpg	0	0	3	0.0%
## 56	hv_0056.jpg	1	0	0	100.0%
## 57	hv_0057.jpg	0	0	3	0.0%
## 58	hv_0058.jpg	0	1	1	0.0%
## 59	hv_0059.jpg	0	0	4	0.0%
## 60	hv_0060.jpg	3	0	1	75.0%
## 61	hv_0061.jpg	1	2	1	25.0%
## 62	hv_0062.jpg	4	2	0	66.7%
## 63	hv_0063.jpg	6	2	0	75.0%
## 64	hv_0064.jpg	4	1	4	44.4%
## 65	hv_0065.jpg	0	2	2	0.0%
## 66	hv_0066.jpg	0	0	0	N/A
## 67	hv_0067.jpg	0	1	2	0.0%
## 68	hv_0068.jpg	0	0	4	0.0%
## 69	hv_0069.jpg	0	0	5	0.0%
## 70	hv_0070.jpg	0	0	2	0.0%
## 71	uniform_0001.jpg	0	0	1	0.0%
## 72	uniform_0002.jpg	0	0	2	0.0%
## 73	uniform_0003.jpg	2	3	0	40.0%
## 74	uniform_0004.jpg	0	0	3	0.0%
## 75	uniform_0005.jpg	1	0	1	50.0%
## 76	uniform_0006.jpg	0	1	1	0.0%
## 77	uniform_0007.jpg	1	2	1	25.0%
## 78	uniform_0008.jpg	0	0	2	0.0%
## 79	uniform_0009.jpg	2	2	2	33.3%
## 80	uniform_0010.jpg	1	2	2	20.0%
## 81	uniform_0011.jpg	2	2	2	33.3%
## 82	uniform_0012.jpg	1	3	3	14.3%
## 83	uniform_0013.jpg	2	4	1	28.6%
## 84	uniform_0014.jpg	1	4	3	12.5%
## 85	uniform_0015.jpg	2	3	3	25.0%

## 86	uniform_0016.jpg	2	1	0	66.7%
## 87	uniform_0017.jpg	2	2	4	25.0%
## 88	uniform_0018.jpg	2	2	6	20.0%
## 89	uniform_0019.jpg	2	1	4	28.6%
## 90	uniform_0020.jpg	5	1	2	62.5%
## 91	uniform_0021.jpg	2	5	3	20.0%
## 92	uniform_0022.jpg	0	0	4	0.0%
## 93	uniform_0023.jpg	1	4	7	8.3%
## 94	uniform_0024.jpg	2	0	3	40.0%
## 95	uniform_0025.jpg	1	2	2	20.0%
## 96	uniform_0026.jpg	1	1	3	20.0%
## 97	uniform_0027.jpg	1	6	0	14.3%
## 98	uniform_0028.jpg	3	4	3	30.0%
## 99	uniform_0029.jpg	5	3	2	50.0%
## 100	uniform_0030.jpg	0	7	2	0.0%
## 101	uniform_0031.jpg	3	5	2	30.0%
## 102	uniform_0032.jpg	1	5	2	12.5%
## 103	uniform_0033.jpg	1	7	0	12.5%
## 104	uniform_0034.jpg	1	8	6	6.7%
## 105	uniform_0035.jpg	2	3	3	25.0%
## 106	uniform_0036.jpg	3	7	3	23.1%
## 107	uniform_0037.jpg	2	4	6	16.7%
## 108	uniform_0038.jpg	2	5	4	18.2%
## 109	uniform_0039.jpg	3	3	4	30.0%
## 110	uniform_0040.jpg	0	7	3	0.0%
## 111	uniform_0041.jpg	0	7	1	0.0%
## 112	uniform_0042.jpg	0	4	3	0.0%
## 113	uniform_0043.jpg	1	0	4	20.0%
## 114	uniform_0044.jpg	1	3	3	14.3%
## 115	uniform_0045.jpg	2	1	2	40.0%
## 116	uniform_0046.jpg	2	2	4	25.0%
## 117	uniform_0047.jpg	0	1	3	0.0%
## 118	uniform_0048.jpg	1	2	2	20.0%
## 119	uniform_0049.jpg	2	3	2	28.6%
## 120	uniform_0050.jpg	0	1	2	0.0%
## 121	uniform_0051.jpg	1	0	1	50.0%
## 122	uniform_0052.jpg	4	1	2	57.1%
## 123	uniform_0053.jpg	1	1	0	50.0%
## 124	uniform_0054.jpg	1	0	2	33.3%
## 125	uniform_0055.jpg	2	0	2	50.0%
## 126	uniform_0056.jpg	0	1	1	0.0%
## 127	uniform_0057.jpg	2	0	5	28.6%
## 128	uniform_0058.jpg	1	0	6	14.3%
## 129	uniform_0059.jpg	0	0	1	0.0%
## 130	uniform_0060.jpg	2	0	1	66.7%
## 131	uniform_0061.jpg	3	0	0	100.0%
## 132	uniform_0062.jpg	2	1	1	50.0%
## 133	uniform_0063.jpg	2	2	0	50.0%
## 134	uniform_0064.jpg	1	5	2	12.5%
## 135	uniform_0065.jpg	4	2	2	50.0%
## 136	uniform_0066.jpg	4	5	3	33.3%
## 137	uniform_0067.jpg	3	0	1	75.0%
## 138	uniform_0068.jpg	1	2	3	16.7%
## 139	uniform_0069.jpg	6	2	3	54.5%

## 140 uniform_0070.jpg	6	2	3	54.5%
## 141 uniform_0071.jpg	3	0	1	75.0%
## 142 uniform_0072.jpg	4	1	2	57.1%
## 143 uniform_0073.jpg	3	1	2	50.0%
## 144 uniform_0074.jpg	3	0	1	75.0%
## 145 uniform_0075.jpg	2	2	3	28.6%
## 146 uniform_0076.jpg	3	1	3	42.9%
## 147 uniform_0077.jpg	2	3	4	22.2%
## 148 uniform_0078.jpg	2	4	4	20.0%
## 149 uniform_0079.jpg	1	3	1	20.0%
## 150 uniform_0080.jpg	2	1	4	28.6%
## 151 uniform_0081.jpg	1	4	2	14.3%
## 152 uniform_0082.jpg	3	2	2	42.9%
## 153 uniform_0083.jpg	3	1	5	33.3%
## 154 uniform_0084.jpg	1	0	3	25.0%
## 155 uniform_0085.jpg	2	0	1	66.7%
## 156 uniform_0086.jpg	2	0	0	100.0%
## 157 uniform_0087.jpg	2	1	0	66.7%
## 158 uniform_0088.jpg	5	1	3	55.6%
## 159 uniform_0089.jpg	2	0	0	100.0%
## 160 uniform_0090.jpg	3	0	0	100.0%
## 161 uniform_0091.jpg	1	2	2	20.0%
## 162 uniform_0092.jpg	1	1	2	25.0%
## 163 uniform_0093.jpg	0	0	4	0.0%
## 164 uniform_0094.jpg	2	0	3	40.0%
## 165 uniform_0095.jpg	0	1	3	0.0%
## 166 uniform_0096.jpg	0	1	2	0.0%
## 167 uniform_0097.jpg	1	1	3	20.0%
## 168 uniform_0098.jpg	1	0	0	100.0%
## 169 uniform_0099.jpg	2	1	2	40.0%
## 170 uniform_0100.jpg	3	1	3	42.9%
## 171 uniform_0101.jpg	0	1	1	0.0%
## 172 uniform_0102.jpg	1	1	2	25.0%
## 173 uniform_0103.jpg	1	2	0	33.3%
## 174 uniform_0104.jpg	1	1	3	20.0%
## 175 uniform_0105.jpg	3	1	1	60.0%
## 176 uniform_0106.jpg	3	0	3	50.0%
## 177 uniform_0107.jpg	2	0	4	33.3%
## 178 uniform_0108.jpg	6	0	0	100.0%
## 179 uniform_0109.jpg	3	2	1	50.0%
## 180 uniform_0110.jpg	2	2	2	33.3%
## 181 uniform_0111.jpg	0	0	3	0.0%
## 182 uniform_0112.jpg	1	1	0	50.0%
## 183 uniform_0113.jpg	4	1	0	80.0%
## 184 uniform_0114.jpg	0	1	1	0.0%
## 185 uniform_0115.jpg	3	0	1	75.0%
## 186 uniform_0116.jpg	1	0	1	50.0%
## 187 uniform_0117.jpg	1	0	2	33.3%
## 188 uniform_0118.jpg	1	3	1	20.0%
## 189 uniform_0119.jpg	3	2	5	30.0%
## 190 uniform_0120.jpg	0	1	4	0.0%
## 191 uniform_0121.jpg	1	10	1	8.3%
## 192 uniform_0122.jpg	0	13	5	0.0%
## 193 uniform_0123.jpg	0	5	1	0.0%

```
## 194 uniform_0124.jpg      0          4          1      0.0%
## 195 uniform_0125.jpg      1         12          1      7.1%
## 196 uniform_0126.jpg      1          3          3     14.3%
## 197 uniform_0127.jpg      0          3          2      0.0%
## 198 uniform_0128.jpg      0          0          4      0.0%
## 199 uniform_0129.jpg      4          2          5     36.4%
## 200 uniform_0130.jpg      1          2          0     33.3%
```

Overall summary matrix:

```
overall_score_stats <- detections %>%
  group_by(annotation) %>%
  summarise(
    n = n(),
    mean = mean(score),
    median = median(score),
    p10 = quantile(score, 0.10),
    p90 = quantile(score, 0.90),
    .groups = "drop"
  ) %>%
  arrange(
    factor(annotation, levels = c("TP", "MIS-ID", "GHOST"))
  )

print(as.data.frame(overall_score_stats), row.names = FALSE)
```

```
##  annotation    n      mean    median      p10      p90
##          TP  346 0.2960081 0.2353087 0.1170687 0.5628277
##        MIS-ID 454 0.2077844 0.1680612 0.1085700 0.3653682
##          GHOST 421 0.1946738 0.1593585 0.1091404 0.3282175
```

Plots

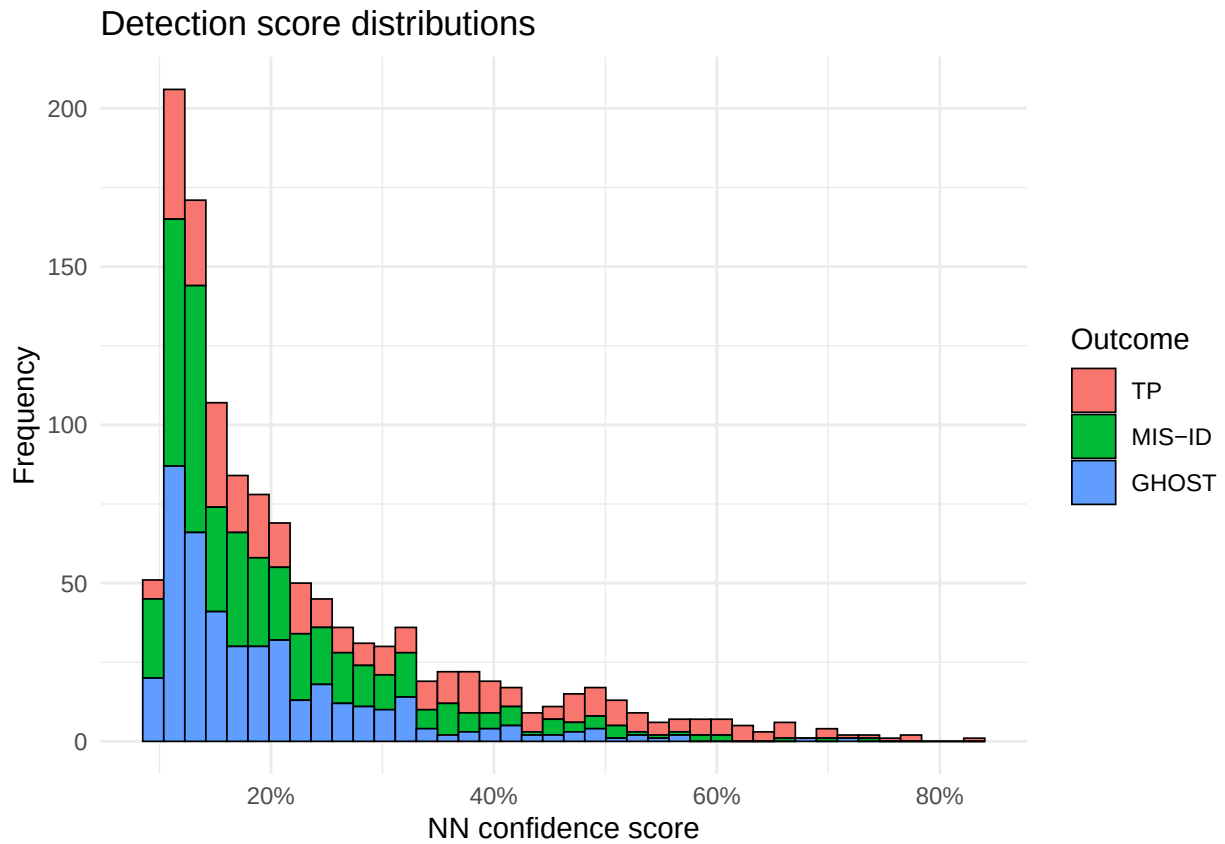
Score distributions:

```
ggplot(
  detections,
  aes(
    x = score,
    fill = annotation
  )
) +
  geom_histogram(
    aes(y = after_stat(count)),
    bins = 40,
    position = "stack",
    colour = "black",
    linewidth = 0.3
  ) +
  scale_x_continuous(
    labels = percent
  ) +
  labs(
    title = "Detection score distributions",
    x = "NN confidence score",
    y = "Frequency",
```

```

    fill = "Outcome"
  ) +
  theme_minimal()

```



Precision vs treshhold:

```

ggplot(
  threshold_stats,
  aes(
    x = threshold,
    y = precision
  )
) +
  geom_line(linewidth = 0.8) +
  geom_point(size = 1.2) +
  scale_x_continuous(
    limits = c(0.10, threshold_max),
    labels = percent
  ) +
  scale_y_continuous(
    labels = percent
  ) +
  coord_cartesian(ylim = c(0, 1)) +
  labs(
    title = "Precision vs confidence threshold",
    x = "Confidence threshold",

```

```

  y = "Precision"
) +
  theme_minimal()

```

```

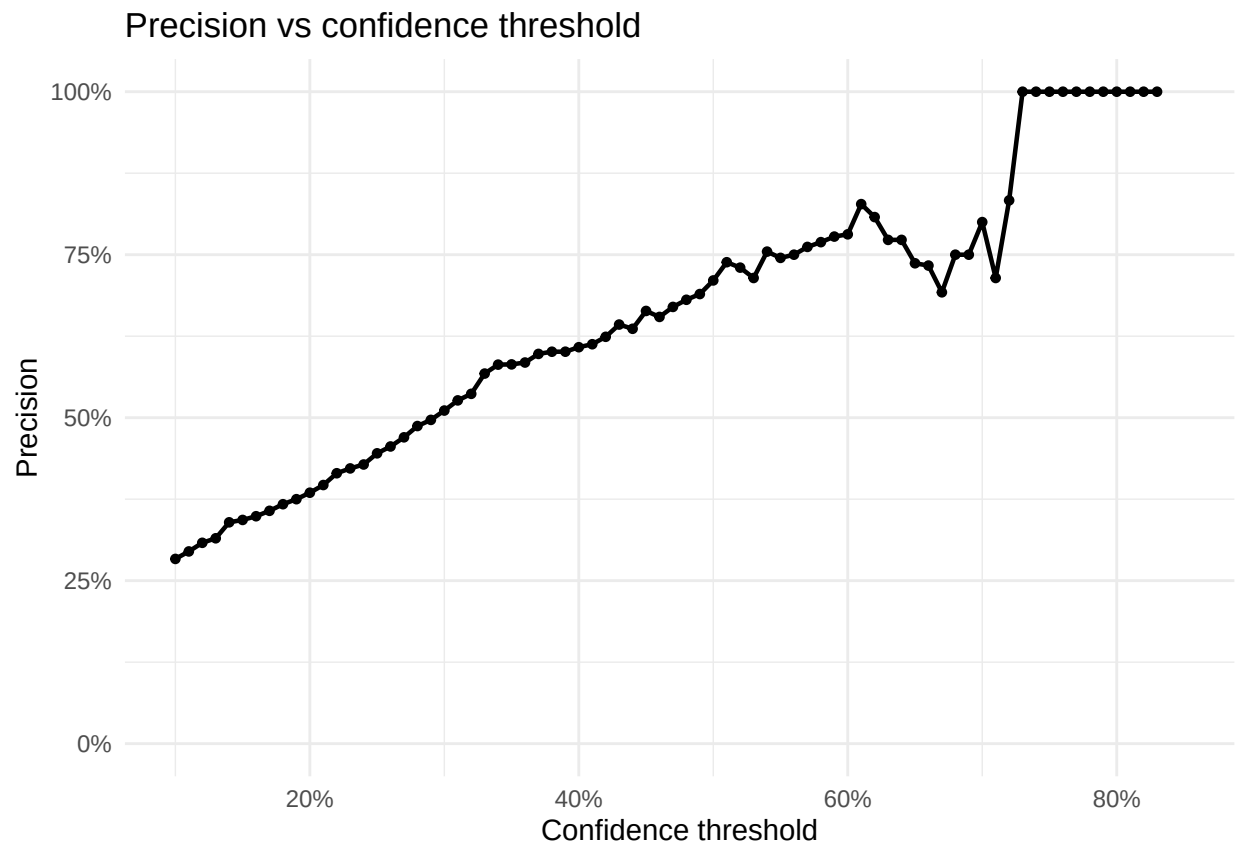
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_line()`).

```

```

## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).

```



Retention vs treshold:

```

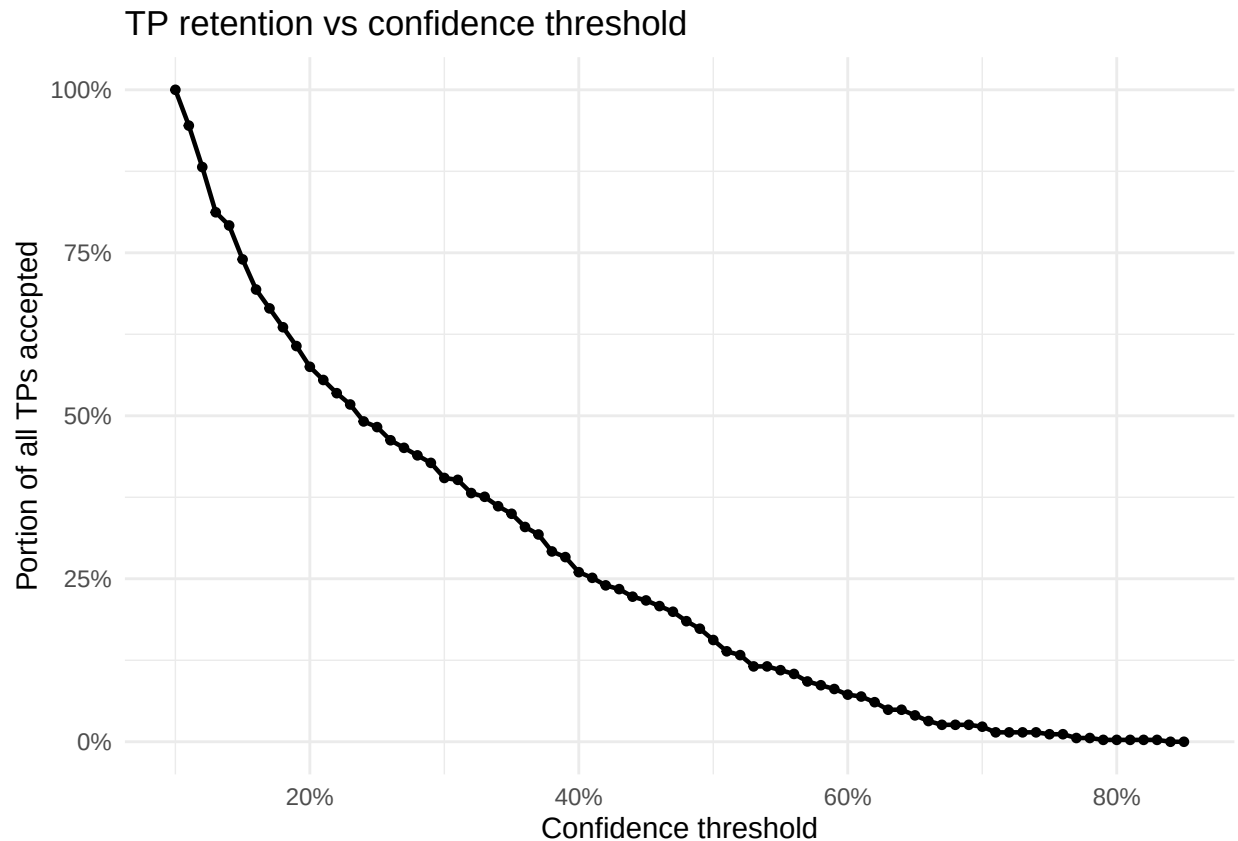
ggplot(
  threshold_stats,
  aes(
    x = threshold,
    y = tp_retention
  )
) +
  geom_line(linewidth = 0.8) +
  geom_point(size = 1.2) +
  scale_x_continuous(
    limits = c(0.10, threshold_max),
    labels = percent
  ) +
  scale_y_continuous(
    limits = c(0, 1),

```

```

labels = percent
) +
labs(
  title = "TP retention vs confidence threshold",
  x = "Confidence threshold",
  y = "Portion of all TPs accepted"
) +
theme_minimal()

```



Interpretation & Threshold selection

Score ranges & TP rate table, seems to confirm that the TP-rate indeed raises with the confidence score, meaning that higher confidence detections can be roughly thought of as “better”

[0.35,0.4) is the first interval where the TP-rate is above 50%, and above that it mostly holds.

This range looks like this in the threshold & precision table:

##	26	0.35	208	121	54	33	58.2%
##	27	0.36	195	114	50	31	58.5%
##	28	0.37	184	110	44	30	59.8%
##	29	0.38	168	101	39	28	60.1%
##	30	0.39	163	98	38	27	60.1%
##	31	0.40	148	90	33	25	60.8%

So a slowly rising precision (58% -> 61%) vs a rapidly falling `n(accepted)` value (208 -> 148). With a 0.10 threshold `n(accepted)` is 1221 so even with a threshold of 0.35 only $\frac{208}{1221} \approx 17,03\%$ of the total classifications would be accepted by the threshold. The lowest threshold where precision is over 50% is 0.30.

Here $\frac{274}{1221} \approx 22,44\%$ are accepted. At $\frac{2}{3}$ precision we're looking at a threshold of 0.47 and an acceptance rate of $\frac{103}{1221} \approx 8,436\%$. Note that this test was done with a 200 frame sample. That'd mean that with a 0.47 threshold we only have $\frac{103}{200} = 0,5150$ accepted detections per frame. With a 0.35 threshold we have $\frac{208}{200} = 1,0400$ accepted detections per frame and with a 0.30 threshold we have $\frac{274}{200} = 1,3700$ accepted detections per frame. Even a 0.30 threshold discards more than half of the detected TPs.

All in all I conclude that this NN is capable of doing the job, but it's expected to do its job fairly poorly. I'd suggest applying a threshold of 0.35 to the NN.